RET fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes RET gene fusions from the msk_impact_50k_2026 study, covering 194 fusion events. 146/194 fusions (75.3%) were in-frame. 179/194 fusions (92.3%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test (p=0.000419666, statistically significant at alpha=0.05). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 75.3% in-frame and 92.3% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 141/146 (96.6%) of in-frame fusions retained the domain, p=0.000419666 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 139/146 (95.2%) of in-frame fusions disrupted the domain, p=0.00178046 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.0348965.

Cutpoint detection

Cutpoint detection scanned 194 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 712 aa (permutation-corrected p=0.000999001, statistically significant at alpha=0.05). This is -12 aa from the nearest configured domain boundary at 724 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=179) and "not_retained" (n=15). For Frame_status, retained: 141/157 (89.8%, 95% CI 84.1%-93.6%); not_retained: 5/14 (35.7%, 95% CI 16.3%-61.2%). Welch's t-test comparing Tumor_variant_count between groups gave means of 67.4413 vs 49.2667, p=0.22222 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (33 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: RET has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIF5B	87	0.4484536082474227	0.30004340774793187	0.14572180003176827	0.9859411065258445	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.4593659412119797	1
CCDC6	51	0.26288659793814434	0.30004340774793187	0.14572180003176827	0.9836137294674462	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.4033467315604365	2
NCOA4	16	0.08247422680412371	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.35137670116522496	3
SPECC1L	4	0.020618556701030927	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.33282000013429713	4
ERCC6	3	0.015463917525773196	0.30004340774793187	0.14572180003176827	1.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3315778680166082	5
TIMM23B	3	0.015463917525773196	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3312736083817198	6
RUFY3	2	0.010309278350515464	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3297272166291425	7
RASSF4	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	1.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3284850845114536	8
SLC4A4	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	1.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3284850845114536	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CCDC186	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	10
CLIP1	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	11
CUBN	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	12
GEMIN5	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	13
GRIPAP1	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	14
KIF13A	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	15
RELCH	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	16
RUFY2	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	17
SHROOM3	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	18
TRIM24	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.9979716024340771	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.32818082487656514	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.920892494929006	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3166189587508045	20
RUFY1	2	0.010309278350515464	0.30004340774793187	0.14572180003176827	0.9006085192697769	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3151227541544974	21
RBPMS	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.8884381338742393	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.31175080459258947	22
ERC1	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.8519269776876268	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.3062741311645976	23
PDCD10	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.8275862068965517	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30262301554593635	24
TFG	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.8275862068965517	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.30262301554593635	25
LIN52	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.7464503042596349	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2904526301503988	26
DOCK1	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.6835699797160244	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2810205814688572	27
KCNMA1	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.6713995943204868	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2791950236595266	28
CTNNA3	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.5496957403651115	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2609394455662203	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1217	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.41784989858012167	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.24116256929847182	30
CTNNA1	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.3772819472616633	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23507737660070308	31
FBXL7	2	0.010309278350515464	0.30004340774793187	0.14572180003176827	0.28803245436105473	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22323634441818913	32
UXS1	1	0.005154639175257732	0.30004340774793187	0.14572180003176827	0.0	0.7278348081553965	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1784850845114536	33

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
219	0	179	1	14	0.0	0.07731958762886598	1.1117104708745449
425	1	178	1	14	0.07865168539325842	0.14903050050745154	0.826724840000421
550	3	176	2	13	0.11079545454545454	0.048844025264081076	1.3111885527954676
556	4	175	2	13	0.14857142857142858	0.06998082470265705	1.1550209437950567
587	5	174	3	12	0.11494252873563218	0.016699929087495975	1.7772853729826312
622	6	173	3	12	0.13872832369942195	0.023865412628667555	1.6222310523597865
627	8	171	5	10	0.0935672514619883	0.001234261100135449	2.9085929583309174
639	9	170	5	10	0.10588235294117647	0.0018331517060542683	2.7368015926613842
640	10	169	5	10	0.11834319526627218	0.0026250181738802633	2.5808676854814183
657	11	168	5	10	0.13095238095238096	0.003644511061609513	2.438360727287861
663	13	166	5	10	0.1566265060240964	0.006510395780065948	2.1863926089934114
671	15	164	5	10	0.18292682926829268	0.01072245099455107	1.9697059299659876
673	16	163	5	10	0.19631901840490798	0.01342338349072885	1.8721380023380016
687	17	162	5	10	0.20987654320987653	0.01656697136370287	1.7807568784395744
708	18	161	5	10	0.2236024844720497	0.020185752752359003	1.694955050558838
712	18	161	14	1	0.007985803016858917	8.448071700649628e-12	11.073242408841473

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
713	176	3	14	1	4.190476190476191	0.27712145920070164	0.5573298428514589
934	176	3	15	0	0.0	1.0	-0.0
1019	177	2	15	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

139	38
7	10

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf00028	PF00028	194	17	1	176	0.003161151466236212

domain_retention tables

frame_domain_contingency_table

141	38
5	10

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	194	179	1	14	0.04988179669030733

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
12	180	194	0.9278350515463918

frequency tables

Partner_gene_counts

Partner_gene	Event_count
CCDC186	1
CCDC6	51
CLIP1	1
CTNNA1	1
CTNNA3	1
CUBN	1
DOCK1	1
EML4	1
ERC1	1
ERCC6	3
FBXL7	2
GEMIN5	1
GRIPAP1	1
KCNMA1	1
KIAA1217	1
KIF13A	1
KIF5B	87
LIN52	1
NCOA4	16
PDCD10	1
RASSF4	1
RBPMS	1
RELCH	1
RUFY1	2
RUFY2	1
RUFY3	2
SHROOM3	1
SLC4A4	1
SPECC1L	4
TFG	1
TIMM23B	3
TRIM24	1
UXS1	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051781-T03-IM6-3	P-0051781-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610377	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0059857-T02-IM7-4	P-0059857-T02-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43611615	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.453+8003:CCDC6_c.2137-417:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0065992-T01-IM7-5	P-0065992-T01-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43611698	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0049461-T02-IM6-6	P-0049461-T02-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611705	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0036379-T01-IM6-7	P-0036379-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611418	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0037412-T01-IM6-8	P-0037412-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610839	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0048611-T01-IM6-9	P-0048611-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611237	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0064853-T01-IM7-10	P-0064853-T01-IM7		CCDC6-RET	CCDC6	out-of-frame	three_prime	43609511	11	627	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv | Protein Fusion: out of frame {CCDC6:RET}	NA
EVT-P-0047144-T01-IM6-11	P-0047144-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611888	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8847:CCDC6_c.2137-144:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038257-T01-IM6-12	P-0038257-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610594	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused with RET exon12):c.304-2131:CCDC6_c.2136+410:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0017365-T04-IM6-13	P-0017365-T04-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0039548-T01-IM6-14	P-0039548-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610843	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0040111-T01-IM6-15	P-0040111-T01-IM6		CCDC6-RET	CCDC6	unknown	three_prime	43609967	11	640	False	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv | Protein Fusion: mid-exon {CCDC6:RET}	NA
EVT-P-0041192-T01-IM6-16	P-0041192-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610571	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T04-IM6-17	P-0022642-T04-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T03-IM6-18	P-0022642-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0051781-T01-IM6-19	P-0051781-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610377	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0042417-T01-IM6-20	P-0042417-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610397	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388:CCDC6_c.2136+213:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004901-T01-IM5-21	P-0004901-T01-IM5		CCDC6-RET	CCDC6	in-frame	three_prime	43611729	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0022380-T03-IM6-22	P-0022380-T03-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43610659	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810:CCDC6_c.2136+475:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0068500-T01-IM7-23	P-0068500-T01-IM7		CCDC6-RET	CCDC6	in-frame	three_prime	43610322	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+11290:CCDC6_c.2136+138:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0057735-T01-IM6-24	P-0057735-T01-IM6		CCDC6-RET	CCDC6	in-frame	three_prime	43611887	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0038053-T01-IM6-26	P-0038053-T01-IM6		CLIP1-RET	CLIP1	unknown	three_prime	43611121	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.21;q24.31)(chr10:g.43611121::chr12:g.122825512) | Protein Fusion: mid-exon {CLIP1:RET}	NA
EVT-P-0000252-T01-IM3-27	P-0000252-T01-IM3		CUBN-RET	CUBN	unknown	three_prime	43610494	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NA | Antisense fusion	NA
EVT-P-0038256-T01-IM6-28	P-0038256-T01-IM6		ERC1-RET	ERC1	unknown	three_prime	43609964	11	639	False	retained	1.0	False	Protein kinase domain	PF00028		ERC1 (NM_178040) - RET (NM_020975) Rearrangement : t(10;12)(q11.1; p13.32)(chr10:g.43609964::chr12:g.1264106) | Protein Fusion: mid-exon {ERC1:RET}	NA
EVT-P-0034781-T01-IM6-29	P-0034781-T01-IM6		GRIPAP1-RET	GRIPAP1	in-frame	three_prime	43611853	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		GRIPAP1 (NM_020137) - RET (NM_020975) rearrangement: t(X;10)(p11.23q11.21)(chrX:g.48831008::chr10:g.43611853) | Protein Fusion: in frame {GRIPAP1:RET}	NA
EVT-P-0026245-T01-IM6-30	P-0026245-T01-IM6		KIAA1217-RET	KIAA1217	unknown	three_prime	43606666	7	425	False	retained	1.0	False	Protein kinase domain	PF00028		KIAA1217 (NM_019590) - RET (NM_020975) fusion (KIAA1217 exons 1-13 fused to RET exons 7-20): c.2390:KIAA1217_c.1275:RETdel | Protein Fusion: mid-exon {KIAA1217:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042153-T01-IM6-31	P-0042153-T01-IM6		KIF13A-RET	KIF13A	in-frame	three_prime	43610510	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF13A (NM_022113) - RET (NM_020975) fusion: t(6;10)(p22.3;q11.21)(chr6:g.17806558::chr10:g.43610510) | Protein Fusion: in frame {KIF13A:RET}	NA
EVT-P-0019535-T02-IM6-32	P-0019535-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611913	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039257-T03-IM7-33	P-0039257-T03-IM7		KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0009514-T01-IM5-34	P-0009514-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611811	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozantinib. PMID: 23533264 | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0055183-T01-IM6-35	P-0055183-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610308	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0011035-T01-IM5-36	P-0011035-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611322	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1726-2296:KIF5B_c.2137-710:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0012648-T01-IM5-37	P-0012648-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611697	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0053175-T01-IM6-38	P-0053175-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611162	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-282:KIF5B_c.2137-870:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052861-T02-IM6-39	P-0052861-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610260	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-20:KIF5B_c.2136+76:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0013825-T01-IM5-40	P-0013825-T01-IM5		KIF5B-RET	KIF5B	unknown	three_prime	43611520	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1782:KIF5B_c.2137-512:RETinv | Protein fusion: mid-exon (KIF5B-RET)	NA
EVT-P-0052826-T01-IM6-41	P-0052826-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611182	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2358:KIF5B_c.2137-850:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0014808-T01-IM6-42	P-0014808-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611055	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0034357-T01-IM6-43	P-0034357-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611187	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0014808-T02-IM6-44	P-0014808-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611056	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0033700-T02-IM6-45	P-0033700-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611670	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0014808-T03-IM6-46	P-0014808-T03-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611056	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0008289-T01-IM5-47	P-0008289-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611748	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000252-T01-IM3-48	P-0000252-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43610597	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RET (NM_020975) Inversion (c.2136+413_KIF5B:c.1726-153inv) | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0008101-T01-IM5-49	P-0008101-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610455	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0048968-T01-IM6-50	P-0048968-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611809	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0048829-T01-IM6-51	P-0048829-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610616	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0065876-T01-IM7-52	P-0065876-T01-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611299	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0016221-T01-IM6-53	P-0016221-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611268	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0037367-T01-IM6-54	P-0037367-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611874	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0057072-T02-IM6-55	P-0057072-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611106	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0017214-T01-IM6-56	P-0017214-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610647	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused in-frame with RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017214-T02-IM6-57	P-0017214-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610647	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused to RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032753-T01-IM6-58	P-0032753-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610925	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1323:KIF5B_c.2136+741:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0057072-T03-IM7-59	P-0057072-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611106	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032686-T01-IM6-60	P-0032686-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611982	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2299:KIF5B_c.2137-50:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0017523-T02-IM6-61	P-0017523-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611192	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2362:KIF5B_c.2137-840:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0046827-T02-IM6-62	P-0046827-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610889	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305:KIF5B_c.2136+705:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032407-T01-IM6-63	P-0032407-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611730	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230:KIF5B_c.2137-302:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0018069-T01-IM6-64	P-0018069-T01-IM6		KIF5B-RET	KIF5B	out-of-frame	five_prime	43607832	8	550	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv | Protein Fusion: out of frame {RET:KIF5B}	NA
EVT-P-0005573-T01-IM5-65	P-0005573-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610418	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1726-860:KIF5B_c.2136+234:RETinv | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004995-T01-IM5-66	P-0004995-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610660	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B(NM_004521) -RET (NM_020975) Fusion (KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+476:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0036092-T01-IM6-67	P-0036092-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610413	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0046726-T01-IM6-68	P-0046726-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610987	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0032227-T01-IM6-69	P-0032227-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611125	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0020273-T02-IM6-70	P-0020273-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611819	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0043007-T02-IM6-71	P-0043007-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610813	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021150-T01-IM6-72	P-0021150-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611591	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021578-T01-IM6-73	P-0021578-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610809	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021879-T01-IM6-74	P-0021879-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610699	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B_c.2136+515_RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021937-T01-IM6-75	P-0021937-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611555	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20) : c.1725+886:KIF5B_c.2137-477:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0021937-T03-IM6-76	P-0021937-T03-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611555	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:KIF5B_c.2137-477:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0043007-T01-IM6-77	P-0043007-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610813	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0042608-T01-IM6-78	P-0042608-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611802	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1670:KIF5B_c.2137-230:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0042580-T01-IM6-79	P-0042580-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610944	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1619:KIF5B_c.2136+760:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0059913-T01-IM7-80	P-0059913-T01-IM7		KIF5B-RET	KIF5B	unknown	three_prime	43610108	11	687	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2441:KIF5B_c.2060:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0001010-T02-IM5-81	P-0001010-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43610863	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-780:KIF5B_c.2136+679:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0060748-T02-IM7-82	P-0060748-T02-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611527	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042104-T01-IM6-83	P-0042104-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610505	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1056:KIF5B_c.2136+321:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0003125-T01-IM5-84	P-0003125-T01-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611584	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NA | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0003125-T01-IM5-85	P-0003125-T01-IM5		KIF5B-RET	KIF5B	in-frame	five_prime	43611581	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		NA | Protein fusion: in frame (RET-KIF5B)	NA
EVT-P-0002873-T01-IM3-86	P-0002873-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43611244	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NA | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0022866-T01-IM6-87	P-0022866-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611871	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_c.2137-161:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0024225-T01-IM6-88	P-0024225-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43609110	10	622	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_c.1866:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0002694-T02-IM5-89	P-0002694-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611682	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NA | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0024843-T01-IM6-90	P-0024843-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611733	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_c.2137-299:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025100-T01-IM6-91	P-0025100-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611511	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2273:KIF5B_c.2137-521:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025100-T02-IM6-92	P-0025100-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611511	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) and RET (NM_020975) Fusion (KIF5B exon 16 fused to RET exon 12): c.1726-2273:KIF5B_c.2137-788:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039941-T01-IM6-93	P-0039941-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611339	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0060748-T03-IM7-94	P-0060748-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43611527	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039474-T02-IM6-95	P-0039474-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610359	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039474-T01-IM6-96	P-0039474-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610359	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0055183-T02-IM6-97	P-0055183-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610308	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025838-T01-IM6-98	P-0025838-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610972	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0025928-T01-IM6-99	P-0025928-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025928-T02-IM6-100	P-0025928-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611910	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0036637-T01-IM6-101	P-0036637-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611965	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0039257-T01-IM6-102	P-0039257-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610060	11	671	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA
EVT-P-0026693-T01-IM6-103	P-0026693-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610511	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion:c.2544+156:KIF5_c.2136+327:RETinv. | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0026777-T01-IM6-104	P-0026777-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0026777-T02-IM6-105	P-0026777-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610717	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0038961-T01-IM6-106	P-0038961-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610775	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0027723-T01-IM6-107	P-0027723-T01-IM6		KIF5B-RET	KIF5B	unknown	three_prime	43610171	11	708	False	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20):c.1725+966:KIF5B_c.2123:RETinv | Protein Fusion: mid-exon {KIF5B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036737-T02-IM6-108	P-0036737-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43612025	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:KIF5B_c.2137-7:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0061331-T01-IM7-109	P-0061331-T01-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610884	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0027991-T01-IM6-110	P-0027991-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43608201	9	550	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0062948-T03-IM7-111	P-0062948-T03-IM7		KIF5B-RET	KIF5B	in-frame	three_prime	43610211	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0064951-T01-IM7-112	P-0064951-T01-IM7		KIF5B-RET	KIF5B	out-of-frame	three_prime	43611130	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv | Protein Fusion: out of frame {KIF5B:RET}	NA
EVT-P-0030453-T01-IM6-113	P-0030453-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43610911	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0037494-T01-IM6-114	P-0037494-T01-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611755	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0000208-T02-IM5-115	P-0000208-T02-IM5		KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12- 20): c.1726-1971:KIF5B_c.2137-385:RETinv | Protein fusion: in frame (KIF5B-RET)	NA
EVT-P-0000208-T01-IM3-116	P-0000208-T01-IM3		KIF5B-RET	KIF5B	in-frame	three_prime	43611647	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv) | Protein fusion: in frame (KIF5B-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016221-T02-IM6-117	P-0016221-T02-IM6		KIF5B-RET	KIF5B	in-frame	three_prime	43611268	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv | Protein Fusion: in frame {KIF5B:RET}	NA
EVT-P-0019663-T01-IM6-118	P-0019663-T01-IM6		LIN52-RET	LIN52	out-of-frame	five_prime	43608613	9	587	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - LIN52 (NM_001024674) Rearrangement : t(10;14)(p32.1;q12)(chr10:g.43608613:chr14:g.74562750) | Protein Fusion: out of frame {RET:LIN52}	NA
EVT-P-0065826-T01-IM7-120	P-0065826-T01-IM7		NCOA4-RET	NCOA4	out-of-frame	three_prime	43610857	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2136+673:RET_c.*23+937NCOA4dup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0037961-T01-IM6-121	P-0037961-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611448	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.763-360:NCOA4_c.2137-584:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0038315-T01-IM6-122	P-0038315-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611981	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0060765-T02-IM7-123	P-0060765-T02-IM7		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611844	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0043077-T01-IM6-124	P-0043077-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611580	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0005748-T01-IM5-125	P-0005748-T01-IM5		NCOA4-RET	NCOA4	in-frame	three_prime	43611325	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.763-260:NCOA4_c.2137-707:RETdup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0047991-T02-IM6-126	P-0047991-T02-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611538	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+537:NCOA4_c.2137-494:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048805-T01-IM6-127	P-0048805-T01-IM6		NCOA4-RET	NCOA4	unknown	three_prime	43611225	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-0049600-T01-IM6-128	P-0049600-T01-IM6		NCOA4-RET	NCOA4	unknown	three_prime	43611613	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-0056870-T01-IM6-129	P-0056870-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43610337	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0052409-T01-IM6-130	P-0052409-T01-IM6		NCOA4-RET	NCOA4	out-of-frame	three_prime	43611691	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0004094-T02-IM6-131	P-0004094-T02-IM6		RELCH-RET	RELCH	in-frame	three_prime	43610279	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		KIAA1468 (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.21;q21.33)(chr10:g.43610279::chr18:g.59909399) | Protein Fusion: in frame {KIAA1468:RET}	NA
EVT-P-0015588-T01-IM6-132	P-0015588-T01-IM6		RET-CCDC186	CCDC186	in-frame	three_prime	43610308	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		C10orf118 (NM_018017) - RET (NM_020975) Rearrangement : c.1326+142:C10orf118_c.2136+124:RETinv | Protein fusion: in frame (C10orf118-RET)	NA
EVT-P-0020974-T01-IM6-133	P-0020974-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611184	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:CCDC6_c.2137-848:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0001745-T02-IM5-134	P-0001745-T02-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611245	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-787:RETinv | Protein fusion: in frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0017365-T02-IM6-135	P-0017365-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0001898-T01-IM3-136	P-0001898-T01-IM3		RET-CCDC6	CCDC6	in-frame	three_prime	43611699	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5 | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0029602-T01-IM6-137	P-0029602-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611160	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0004916-T01-IM5-138	P-0004916-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6(NM_005436)-RET(NM_020975) Fusion(CCDC6 exon 2 fused to RET exon11) : c.304-1153:CCDC6_c.2136+169:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0000425-T01-IM3-139	P-0000425-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43610450	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv) | Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-0031330-T01-IM6-140	P-0031330-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610353	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+3703:CCDC6_c.2136+169:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0030990-T03-IM6-141	P-0030990-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611881	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-151:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0005168-T01-IM5-142	P-0005168-T01-IM5		RET-CCDC6	CCDC6	out-of-frame	three_prime	43608797	10	587	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 Exon1 fused to RET Exon10) : c.304-11493:CCDC6_c.1760-207:RETinv | Protein fusion: out of frame (CCDC6-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0022380-T01-IM6-143	P-0022380-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610659	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022380-T02-IM6-144	P-0022380-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610659	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET(NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475:c.2136+475inv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0029315-T02-IM6-145	P-0029315-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610631	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0029315-T01-IM6-146	P-0029315-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610631	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20: c.304-20462:CCDC6_c.2136+447:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0000889-T01-IM3-147	P-0000889-T01-IM3		RET-CCDC6	CCDC6	in-frame	five_prime	43611994	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv | Protein fusion: in frame (RET-CCDC6)	NA
EVT-P-0007322-T02-IM6-148	P-0007322-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610268	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0027953-T01-IM6-149	P-0027953-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611278	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0008504-T03-IM6-150	P-0008504-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610583	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008955-T03-IM5-151	P-0008955-T03-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611019	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0013023-T01-IM5-152	P-0013023-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611810	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0013394-T01-IM5-153	P-0013394-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43610232	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0022607-T01-IM6-154	P-0022607-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611212	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12 - 20): c.304-414:CCDC6_c.2137-820:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T01-IM6-155	P-0022642-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0022642-T02-IM6-156	P-0022642-T02-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611768	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0017365-T03-IM6-157	P-0017365-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611464	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA
EVT-P-0025349-T03-IM6-158	P-0025349-T03-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43611986	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv | Protein Fusion: in frame {CCDC6:RET}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049588-T01-IM6-159	P-0049588-T01-IM6		RET-CCDC6	CCDC6	out-of-frame	five_prime	43609437	10	627	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-15222:CCDC6inv | Protein Fusion: out of frame {RET:CCDC6}	NA
EVT-P-0015435-T01-IM6-160	P-0015435-T01-IM6		RET-CCDC6	CCDC6	in-frame	three_prime	43610969	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0005556-T01-IM5-161	P-0005556-T01-IM5		RET-CCDC6	CCDC6	in-frame	three_prime	43611828	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv | Protein fusion: in frame (CCDC6-RET)	NA
EVT-P-0063502-T01-IM7-162	P-0063502-T01-IM7		RET-CTNNA1	CTNNA1	unknown	five_prime	43622038	19	1019	False	retained	1.0	False	Protein kinase domain; PF00028			RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827::chr10:g.43622038) | Protein Fusion: mid-exon {RET:CTNNA1}	NA
EVT-P-0063802-T01-IM7-163	P-0063802-T01-IM7		RET-CTNNA3	CTNNA3	out-of-frame	five_prime	43617566	16	934	True	disrupted	0.74822695035461	True	PF00028		Protein kinase domain	RET (NM_020975) - CTNNA3 (NM_013266) rearrangement: c.2801+102:RET_c.1375-34298:CTNNA3inv | Protein Fusion: out of frame {RET:CTNNA3}	NA
EVT-P-0030396-T02-IM6-164	P-0030396-T02-IM6		RET-DOCK1	DOCK1	unknown	three_prime	43608320	9	556	False	retained	1.0	False	Protein kinase domain	PF00028		DOCK1 (NM_001380) - RET (NM_020975)Rearrangement: c.2515+295:RET_c.1668:RET dup | Protein Fusion: mid-exon {DOCK1:RET}	NA
EVT-P-0013886-T01-IM5-165	P-0013886-T01-IM5		RET-EML4	EML4	unknown	three_prime	43610067	11	673	False	retained	1.0	False	Protein kinase domain	PF00028		EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.43610067) | Protein fusion: mid-exon (EML4-RET)	NA
EVT-P-0014808-T03-IM6-166	P-0014808-T03-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv | Protein Fusion: out of frame {RET:ERCC6}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014808-T02-IM6-167	P-0014808-T02-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv | Protein Fusion: out of frame {RET:ERCC6}	NA
EVT-P-0014808-T01-IM6-168	P-0014808-T01-IM6		RET-ERCC6	ERCC6	out-of-frame	five_prime	43611216	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6inv | Protein fusion: out of frame (RET-ERCC6)	NA
EVT-P-0045809-T02-IM6-169	P-0045809-T02-IM6		RET-FBXL7	FBXL7	in-frame	five_prime	43622269	19	1063	True	retained	1.0	False	Protein kinase domain; PF00028			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622269) | Protein Fusion: in frame {RET:FBXL7}	NA
EVT-P-0045809-T06-IM6-170	P-0045809-T06-IM6		RET-FBXL7	FBXL7	in-frame	five_prime	43622323	19	1063	True	retained	1.0	False	Protein kinase domain; PF00028			RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622323) | Protein Fusion: in frame {RET:FBXL7}	NA
EVT-P-0025215-T01-IM6-171	P-0025215-T01-IM6		RET-GEMIN5	GEMIN5	in-frame	three_prime	43611067	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		GEMIN5 (NM_015465) - RET (NM_020975) rearrangement: t(5;10)(q33.2;q11.21)(chr5:g.154274993::chr10:g.43611067) | Protein Fusion: in frame {GEMIN5:RET}	NA
EVT-P-0027417-T01-IM6-173	P-0027417-T01-IM6		RET-KCNMA1	KCNMA1	out-of-frame	three_prime	43608158	9	550	True	retained	1.0	False	Protein kinase domain	PF00028		KCNMA1 (NM_001161352) - RET (NM_020975) rearrangement: c.378+15627:KCNMA1_c.1649-143:RET | Protein Fusion: out of frame {KCNMA1:RET}	NA
EVT-P-0043704-T01-IM6-174	P-0043704-T01-IM6		RET-KIF5B	KIF5B	in-frame	five_prime	43610331	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5Binv | Protein Fusion: in frame {RET:KIF5B}	NA
EVT-P-0006226-T02-IM5-178	P-0006226-T02-IM5		RET-NCOA4	NCOA4	in-frame	three_prime	43611893	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0001151-T01-IM3-179	P-0001151-T01-IM3		RET-NCOA4	NCOA4	in-frame	three_prime	43611645	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2137-387_c.1840-75dup | Protein fusion: in frame (NCOA4-RET)	NA

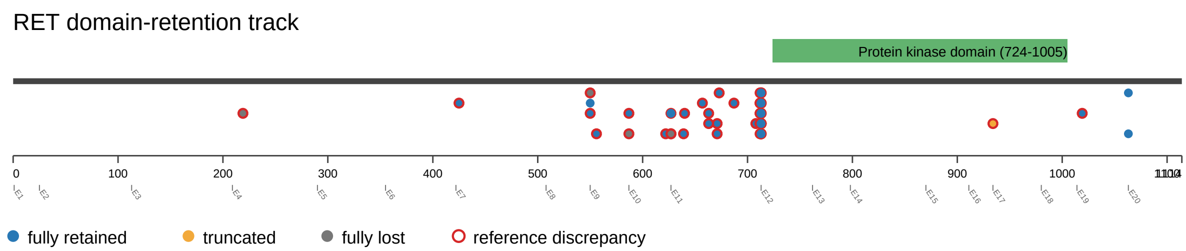
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009592-T01-IM5-180	P-0009592-T01-IM5		RET-NCOA4	NCOA4	unknown	three_prime	43611143	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET exons 12=20: c.872:NCOA4_c.2136+621:RETdup | Protein fusion: mid-exon (NCOA4-RET)	NA
EVT-P-0001155-T01-IM3-181	P-0001155-T01-IM3		RET-NCOA4	NCOA4	in-frame	three_prime	43610941	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2136+757_c.715-617dup | Protein fusion: in frame (NCOA4-RET)	NA
EVT-P-0022818-T01-IM6-182	P-0022818-T01-IM6		RET-NCOA4	NCOA4	unknown	three_prime	43611746	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup | Protein Fusion: mid-exon {NCOA4:RET}	NA
EVT-P-0002145-T01-IM3-183	P-0002145-T01-IM3		RET-PDCD10	PDCD10	out-of-frame	five_prime	43609394	10	627	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - PDCD10 (NM_145860) translocation: t(10,3)(q11.21;q26.1)(chr10:g.43609394::chr3:g.167419184) | Protein fusion: out of frame (RET-PDCD10)	NA
EVT-P-0001898-T01-IM3-184	P-0001898-T01-IM3		RET-RASSF4	RASSF4	in-frame	five_prime	43611749	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - RASSF4 (NM_032023) deletion: c.2137-283_c.138+357del | Protein fusion: in frame (RET-RASSF4)	NA
EVT-P-0030396-T02-IM6-185	P-0030396-T02-IM6		RET-RBPMS	RBPMS	unknown	three_prime	43610018	11	657	False	retained	1.0	False	Protein kinase domain	PF00028		RBPMS (NM_001008712) - RET (NM_020975) Rearrangement : t(8,10)(p21.1,q11.21)(chr8:g.30412656::chr10:g.43610018) | Protein Fusion: mid-exon {RBPMS:RET}	NA
EVT-P-0029222-T02-IM6-212	P-0029222-T02-IM6		RET-RUFY1	RUFY1	unknown	three_prime	43610037	11	663	False	retained	1.0	False	Protein kinase domain	PF00028		RUFY1 (NM_025158) - RET (NM_020975) Rearrangement : t(5,10)(q35.3,q11.21)(chr5:g.179025890::chr10:g.43610037) | Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-0015310-T02-IM6-213	P-0015310-T02-IM6		RET-RUFY3	RUFY3	out-of-frame	three_prime	43610427	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655316::chr10:g.43610427) | Protein fusion: out of frame (RUFY3-RET)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0015310-T04-IM6-214	P-0015310-T04-IM6		RET-RUFY3	RUFY3	unknown	three_prime	43610427	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655251::chr10:g.43610427) | Protein Fusion: mid-exon {RUFY3:RET}	NA
EVT-P-0025609-T01-IM6-215	P-0025609-T01-IM6		RET-SHROOM3	SHROOM3	in-frame	three_prime	43611545	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		SHROOM3 (NM_020859) - RET (NM_020975) rearrangement: t(4;10)(q21.1;q11.21)(chr4:g.77664882::chr10:g.43611545) | Protein Fusion: in frame {SHROOM3:RET}	NA
EVT-P-0053530-T01-IM6-216	P-0053530-T01-IM6		RET-SLC4A4	SLC4A4	out-of-frame	five_prime	43610439	11	712	True	lost	0.0	False	PF00028	Protein kinase domain		RET (NM_020975) - SLC4A4 (NM_001134742) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.72412310::chr10:g.43610439) | Protein Fusion: out of frame {RET:SLC4A4}	NA
EVT-P-0002145-T01-IM3-217	P-0002145-T01-IM3		RET-TFG	TFG	in-frame	three_prime	43609437	11	627	True	retained	1.0	False	Protein kinase domain	PF00028		RET (NM_020975) - TFG (NM_001195478) translocation: t(10,3)(q11.21;q12.2)(chr10:g.43609437::chr3:g.100450751) | Protein fusion: in frame (TFG-RET)	NA
EVT-P-0021980-T01-IM6-218	P-0021980-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43610550	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused to RET exons 12- 20): c.763-591:NCOA4_c.2136+366:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0017716-T01-IM6-219	P-0017716-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43611172	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-7 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0025531-T01-IM6-220	P-0025531-T01-IM6		RET-TIMM23B	TIMM23B	out-of-frame	three_prime	43611375	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		NCOA4 (NM_001145260) RET (NM_020975) fusion: c.619-108NCOA4_c.2137-657RETdup | Protein Fusion: out of frame {TIMM23B:RET}	NA
EVT-P-0034704-T01-IM6-222	P-0034704-T01-IM6		RET-TRIM24	TRIM24	in-frame	three_prime	43611827	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		TRIM24 (NM_015905) - RET (NM_020975) Fusion (TRIM24 exon 9 fused with RET exon12) : t(7;10)(q34;q11.21)(chr7:g.138243011::chr10:g.43611827) | Protein Fusion: in frame {TRIM24:RET}	NA

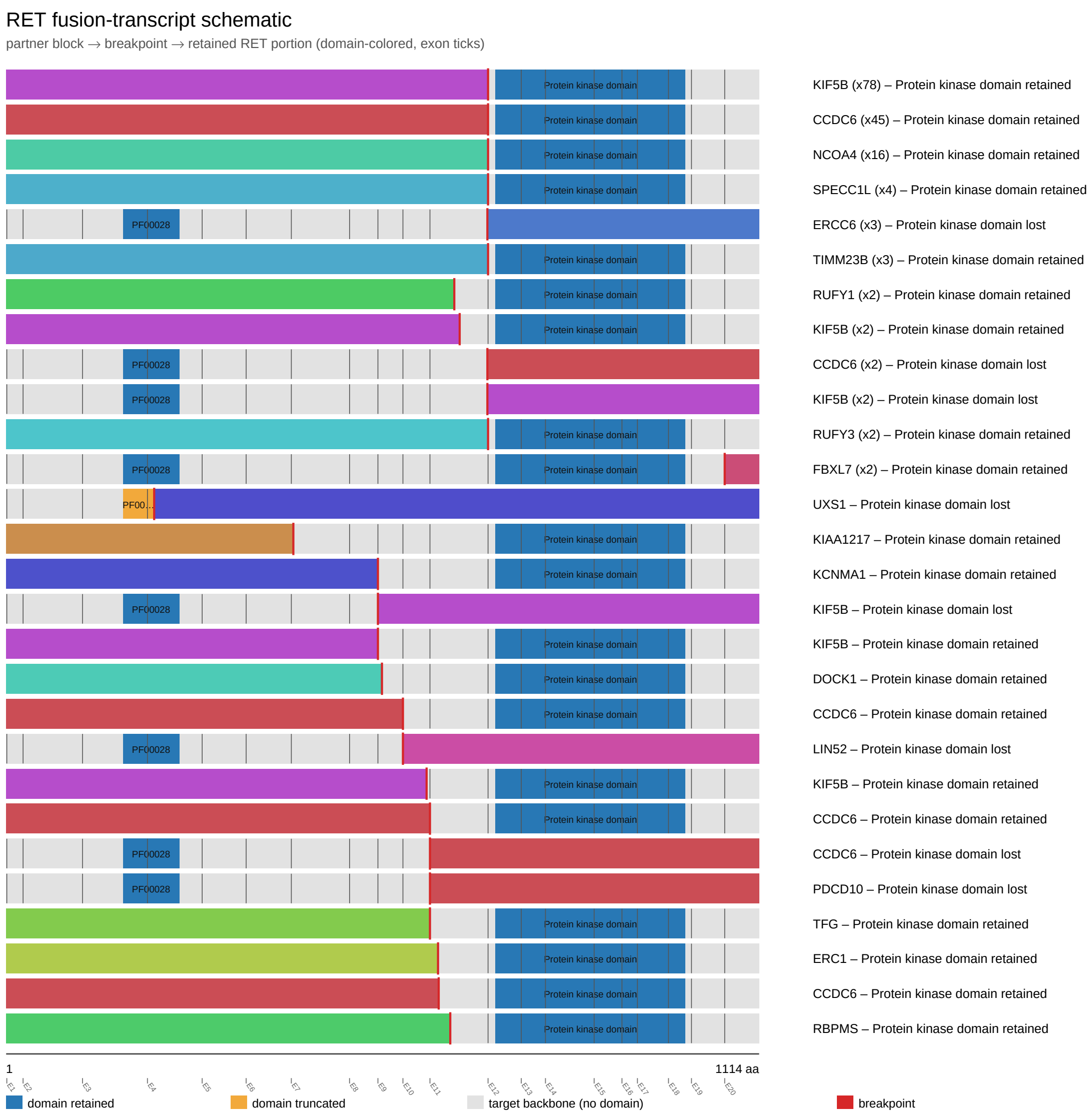
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0046790-T01-IM6-223	P-0046790-T01-IM6		RET-UXS1	UXS1	unknown	five_prime	43600430	4	219	False	lost	0.0	False		Protein kinase domain	PF00028	RET (NM_020975) - UXS1 (NM_001253875) rearrangement: t(2;10)(q12.2;q11.21)(chr2:g.106728260::chr10:g.43600430) | Protein Fusion: mid-exon {RET:UXS1}	NA
EVT-P-0029222-T03-IM6-224	P-0029222-T03-IM6		RUFY1-RET	RUFY1	unknown	three_prime	43610037	11	663	False	retained	1.0	False	Protein kinase domain	PF00028		RUFY1 (NM_025158) - RET (NM_020975) fusion: t(5;10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037) | Protein Fusion: mid-exon {RUFY1:RET}	NA
EVT-P-0031340-T02-IM6-225	P-0031340-T02-IM6		RUFY2-RET	RUFY2	in-frame	three_prime	43611559	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		RUFY2 (NM_017987) - RET (NM_020975) fusion: c.1045-640:RUFY2_c.2137-473:RETinv | Protein Fusion: in frame {RUFY2:RET}	NA
EVT-P-0026614-T01-IM6-227	P-0026614-T01-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43611575	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(10q11.21;22q11.23)(chr10:g.43611575::chr22:g.24740413) | Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-0036400-T06-IM6-228	P-0036400-T06-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		SPECC1L (NM_001145468) - RET (NM_020975) fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-0036400-T04-IM6-229	P-0036400-T04-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA
EVT-P-0036400-T03-IM6-230	P-0036400-T03-IM6		SPECC1L-RET	SPECC1L	in-frame	three_prime	43610353	12	713	True	retained	1.0	False	Protein kinase domain	PF00028		SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056) | Protein Fusion: in frame {SPECC1L:RET}	NA

Figures

domain retention outliers



fusion schematic



Showing the top 28 of 45 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

reference comparison

Reference vs msk_impact_50k_2026